

Research Objectives

- **Develop a Hybrid SLAM Navigation System:** Design an autonomous navigation and positioning system for greenhouse mobile robots that combines 3D Lidar spatial sensing with 2D Lidar SLAM processing in Cartographer.
- **Reduce Computational Burden:** Filter and fuse 3D point cloud data collected from a multi-line Lidar into 2D laser data to lower the computing requirements on industrial PCs while preserving critical 3D environment obstacle details.
- **Optimize Path Planning:** Utilize the Dijkstra algorithm for global path planning and the Dynamic Window Approach (DWA) algorithm for local real-time obstacle avoidance in greenhouse crop rows.
- **Verify System Precision:** Validate the positioning accuracy, lateral deviation, and heading/course deviation of the mobile robot across multiple operating speeds in an actual greenhouse setting.

Research Gaps

- **Limitations of Standalone 2D Lidar SLAM:** Standard 2D Lidar only senses obstacles on a single horizontal plane, leaving hazards above or below the sensor height undetected in complex greenhouse environments.
- **High Computational Demand of Pure 3D Lidar SLAM:** Processing raw, uncompressed 3D point cloud data continuously requires high-end, expensive industrial computers with large power consumption.
- **Inapplicability of Satellite Navigation (GNSS):** Global Navigation Satellite Systems (GNSS) fail inside greenhouse structures due to signal obstruction by roofs, walls, and dense plant canopies.
- **Limitations of Fixed-Route Navigation:** Traditional guide rails, magnetic tape, and ribbon navigation methods impose rigid movement paths, occupy valuable ground space, and incur high installation/maintenance costs.
- **Sensitivity of Visual SLAM:** Machine vision and vision-based SLAM systems are vulnerable to unpredictable lighting and shadow fluctuations common in greenhouses.

Research Questions

- **Data Reduction & Fusion:** How effectively can 3D Lidar point clouds be filtered and compressed into 2D LaserScan topics without losing critical spatial obstacle information needed for safe navigation?
- **Localization Accuracy:** What level of positioning accuracy (cm) and heading angle precision (degrees) can be achieved using the Adaptive Monte Carlo Localization (AMCL) algorithm combined with wheel encoders and an IMU?

- **Speed vs. Path Tracking Performance:** How do varying operational speeds ($0.2m/s$, $0.4m/s$, and $0.6m/s$) impact the robot's lateral and course deviations during global and local path execution?
- **Greenhouse Feasibility:** Can the proposed hybrid system reliably operate within narrow greenhouse crop rows (e.g., $1m$ row spacing) without colliding with cultivation tanks or plant canopies?

Title: Autonomous Systems: Indoor Drone Navigation

Research Objectives

- **Extend Nav2 Framework to UAVs:** Adapt the ROS Navigation2 (Nav2) stack and SLAM Toolbox—originally built for 2D terrestrial ground robots—to enable autonomous indoor flight for Unmanned Aerial Vehicles (UAVs).
- **Achieve GPS-Denied Autonomous Navigation:** Develop a simulated quadcopter platform capable of autonomous mapping, localization, path planning, and obstacle avoidance in indoor environments where GPS signals are unavailable.
- **Dimensionality Reduction for Flight Control:** Implement a stabilization control mechanism using PID controllers to keep the quadcopter hovering at a fixed altitude (z -axis), reducing the 3D flight problem into a manageable 2D navigation plane.
- **Map Ground Interfaces to UAV Motion: Establish a 1:1** mapping between standard ROS Twist velocity commands (used by ground robots like TurtleBots) and the quadcopter's flight dynamics (pitch, roll, yaw, and thrust).

Research Gaps

- **Lack of UAV Support in Nav2:** Existing ROS 2 navigation frameworks (Nav2) are designed primarily for land-based mobile robots; their application and extension to aerial vehicles remain underdeveloped.
- **Ineffectiveness of Wheel-Based Odometry:** Standard mobile robot odometry relies on wheel radius and RPM computations, which cannot be applied to drones driven by propellers.
- **GPS Signal Obstruction Indoors:** Reinforced concrete structures and indoor obstacles block satellite signals, rendering traditional GNSS-based navigation unusable for indoor drones.
- **Payload and Compute Constraints:** Small indoor UAVs face strict size, weight, and battery constraints, limiting their ability to carry heavy processing hardware or complex sensor suites.

Research Questions

- **Nav2 Adaptability:** How effectively can the Nav2 framework and 2D SLAM Toolbox generate costmaps and collision-free trajectories for a 3D flying quadcopter when mapped through a 2D hover plane?
- **3D Laser Odometry Estimation:** Can the rf2o_laser_odometry package reliably estimate a drone's real-time position and orientation in the global reference frame using 2D/3D Lidar scan data without relying on physical wheel encoders?
- **Fixed-Altitude Flight Control:** Does a PID controller regulating thrust, roll, pitch, and yaw provide sufficient hover stability to treat horizontal maneuvers (x, y) as simple 2D planar navigation?
- **Feasibility of Standard Interfaces:** Can a quadcopter be controlled seamlessly by translating standard cmd_vel (Twist) messages into joint velocity inputs via ros2_control without modifying the core Nav2 architecture?

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Title: A Layered ROS 2/Nav2 Architecture for Fail-Safe Indoor UAV Navigation with Deterministic Dynamic-Obstacle Braking

Goal: Design, implement, and evaluate an indoor multirotor autonomy stack—using ROS 2 Jazzy, Gazebo Harmonic, ArduPilot Copter SITL, SLAM Toolbox, and Nav2—that decouples emergency collision intervention from normal planning/costmap latency using an independent Nav2 Collision Monitor layer to safely navigate narrow indoor corridors with dynamic obstacles.

Research Objectives

- **Layered Safety Architecture:** Separate global path planning, local trajectory tracking, and emergency collision intervention into distinct layers to isolate emergency braking from planner/costmap latency.
- **Nav2 Adaptation for Aerial Platforms:** Adapt the ROS 2 Jazzy Nav2 stack and SLAM Toolbox to work alongside ArduPilot Copter SITL (in GUIDED mode) for autonomous indoor UAV navigation.
- **Planner-Agnostic Dynamic Obstacle Braking:** Deploy Nav2 Collision Monitor to directly inspect raw LiDAR scans and immediately clamp incoming velocity commands to zero upon obstacle detection before reaching the flight controller.
- **Staged Validation Framework:** Evaluate the integrated software stack across an eleven-stage progressive validation ladder to systematically verify flight stability, mapping, path tracking, and safety responses.

Research Gaps

- **High-Latency Emergency Response in Aerial Nav2:** Standard Nav2 pipeline execution rely on costmap updates and planner replanning cycles, which introduce latencies unsuitable for immediate quadrotor emergency stopping.

- **Aerial Attitude and Deceleration Dynamics:** Ground-robot navigation assumptions ignore that drones cannot stop instantaneously and require time and attitude adjustments to dissipate forward momentum.
- **Ground-Plane Sensor Artifacts on Aerial Platforms:** Aerial hover pitch oscillations cause depth cameras to capture floor returns, creating artificial lethal costmap obstacles if parameters are configured with standard ground-robot thresholds.
- **Simulation-to-Real Deployment Gaps:** Lack of hardware-in-the-loop (HITL) or physical flight testing to account for physical aerodynamic phenomena like rotor wash, ground effects, and communication packet drops.

Research Questions

- **Intervention Latency Reduction:** Can positioning Nav2 Collision Monitor directly between the navigation command path and flight-control bridge achieve deterministic, low-latency velocity clamping without relying on replanning cycles?
- **3D Altitude/Attitude Stability:** Does a 2D circular StopZone (0.80 m radius) provide sufficient spatial clearance and time for physical drone deceleration ($< 1.5s$) in narrow corridor environments?
- **Costmap False-Positive Mitigation:** How can depth-camera minimum-obstacle-height thresholds be configured to prevent floor returns caused by pitch jitter while retaining detection of actual obstacles?
- **Command Translation Reliability:** Can a custom interface node (`cmd_vel_bridge`) reliably convert standard ROS 2 `geometry_msgs/Twist` commands to `TwistStamped` for ArduPilot external velocity control without causing timeouts or control exceptions?